Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Mechanics 1 – Elastic Strings & Springs

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. State Hooke's law for an elastic string of natural length l and modulus λ. **[2]**

2. An elastic string of natural length 2 m and modulus 20 N is stretched to 2.5 m. Find the tension. **[3]**

3. Find the elastic potential energy stored in the string in the previous question. **[3]**

4. A spring of natural length 0.5 m and modulus 10 N is compressed by 0.1 m. Find the thrust and EPE. **[4]**

5. A particle of mass 0.5 kg hangs in equilibrium on an elastic string of natural length 1 m and modulus 19.6 N. Find the extension. **[4]**

6. A particle is attached to two elastic strings on a smooth horizontal table. Describe the equilibrium position. **[2]**

7. State the formula for elastic potential energy. **[2]**

8. An elastic string of natural length l and modulus mg supports a mass m in equilibrium. Find the extension. **[3]**

9. A particle of mass m is attached to a fixed point by an elastic string of natural length l and modulus λ. It hangs in equilibrium then is pulled down a further distance a and released. Write the energy equation at extension x. **[4]**

10. Find the speed of the particle in the previous question when the string first becomes slack (if it does). **[4]**

11. A spring has stiffness $k = 50\,\text{N}\,\text{m}^{-1}$. Find the force when compressed by 4 cm. **[2]**

12. Distinguish between modulus of elasticity and stiffness. **[2]**

13. An elastic string stretches by 5 cm under a load of 2 N. Find λ if natural length is 40 cm. **[3]**

14. A particle oscillates on a smooth table between two points where the string is taut. Describe the motion qualitatively. **[3]**

15. Find the work done in stretching an elastic string from natural length l to length $l + e$. **[3]**

Section B – Hard (Q16–25)

16. A particle of mass 2 kg is attached to a fixed point by an elastic string of natural length 1 m and modulus 40 N. It is released from rest at the level of the fixed point with the string horizontal and just taut (on a smooth table). Find the greatest distance from the fixed point in the subsequent motion. **[6]**

17. A mass m hangs on an elastic string of natural length l, modulus $2mg$. It is pulled down a distance $\frac{l}{2}$ from equilibrium and released. Find the maximum speed. **[6]**

18. Two elastic strings, each of natural length 0.5 m and modulus 10 N, are attached to fixed points A and B distance 2 m apart on a smooth table, and to a particle P of mass 0.2 kg. Find the period of small oscillations along AB. **[6]**

19. A particle performs SHM while attached to an elastic string. Show that the period is $2\pi\sqrt{\frac{ml}{\lambda}}$. **[4]**

20. An elastic string of natural length 2 m and modulus 30 N has one end fixed and a mass 1 kg on the other. The mass is held with the string horizontal and just taut, then released (smooth table). Find the speed when the string is vertical (if constrained) or discuss. **[5]**

21. A spring is compressed by x against a mass m on a smooth surface then released. Find the speed when the spring reaches natural length. **[4]**

22. A particle slides down a smooth incline of angle α, connected to the top by an elastic string. Set up the energy equation. **[5]**

23. Find the extension in equilibrium when a mass m hangs from two parallel identical elastic strings. **[3]**

24. An elastic string is stretched from l to $2l$. Show the EPE is $\frac{\lambda l}{2}$. **[3]**

25. A bungee jumper of mass m falls from rest attached to a cord of natural length l and modulus kmg. Find the maximum extension. **[6]**

End of Worksheet • *Show all working clearly. Take* $g = 9.8\,\mathrm{m\,s^{-2}}$ *unless otherwise stated.*